Work-energy lab

antocciphysics | TEACHER KEY | CPA

Learning target: Compare measured work with kinetic-energy change.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 3; 3 course days later.*
-15 J; the force opposes displacement.

02. Explain a worked example (worked)
Work 12 J versus energy change 11 J; check force calibration, distance, speed or omitted friction.

03. Look and explain (model)
6 J.

04. Build the solution (practice)
K 1 J→4 J;ΔK 3 J; matches 3 J network.

05. Use the idea (reason)
Units, force calibration, friction, velocity interval, moving pulley or other energy; preserve actual data.

06. Choose a method (mixed) *Review: Unit 3, Day 8; 12 course days later.*
0 m/s; equal opposite momenta cancel, and the combined mass is 8 kg.

07. Exit check (exit)
The energy theorem compares the same interval; mismatched intervals need not agree.

08. STRETCH: Apply it another way (stretch)
1±2 contains zero; 5±2 does not. Neither result alone identifies the cause.

Teaching note: Investigation companion. Use only teacher-approved equipment and label supplied versus measured data.

Work-energy lab

antocciphysics | TEACHER KEY | HONORS

Learning target: Compare measured work with kinetic-energy change.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 3; 3 course days later.*
W=-15 J because cos 120°=-1/2; the component opposes displacement.

02. Explain a worked example (worked)
Work 12 J versus energy change 11 J; check force calibration, distance, speed or omitted friction.

03. Represent and interpret (model)
6 J.

04. Solve and justify (practice)
ΔK=.25(1.44-.16)=.32 J; resistance work≈-.03 J.

05. Analyze the reasoning (reason)
Units, force calibration, friction, velocity interval, moving pulley or other energy; preserve actual data.

06. Choose a method (mixed) *Review: Unit 3, Day 8; 12 course days later.*
v_f=6 m/s. Initial K=720 J; final K=72 J; fraction 0.10 or 10%.

07. Exit check (exit)
Work 2 J; speed 2 m/s.

08. STRETCH: Apply it another way (stretch)
ΔK=8 J; friction work -2 J; mean friction magnitude 1 N.

Teaching note: Investigation companion. Use only teacher-approved equipment and label supplied versus measured data.